

NovaDiff — Target codebase

Baseline: f7c2bfd81af2

Target: NovaDiff

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Diagrams use Mermaid heuristics (imports, cross-file call guesses, type sketches). They are **not** compiler-grade call graphs. The module graph is file-to-file (JS/TS relative imports). The call graph links files when a callee name maps to exactly one definition in the repo (regex across many languages).

AI target codebase narrative

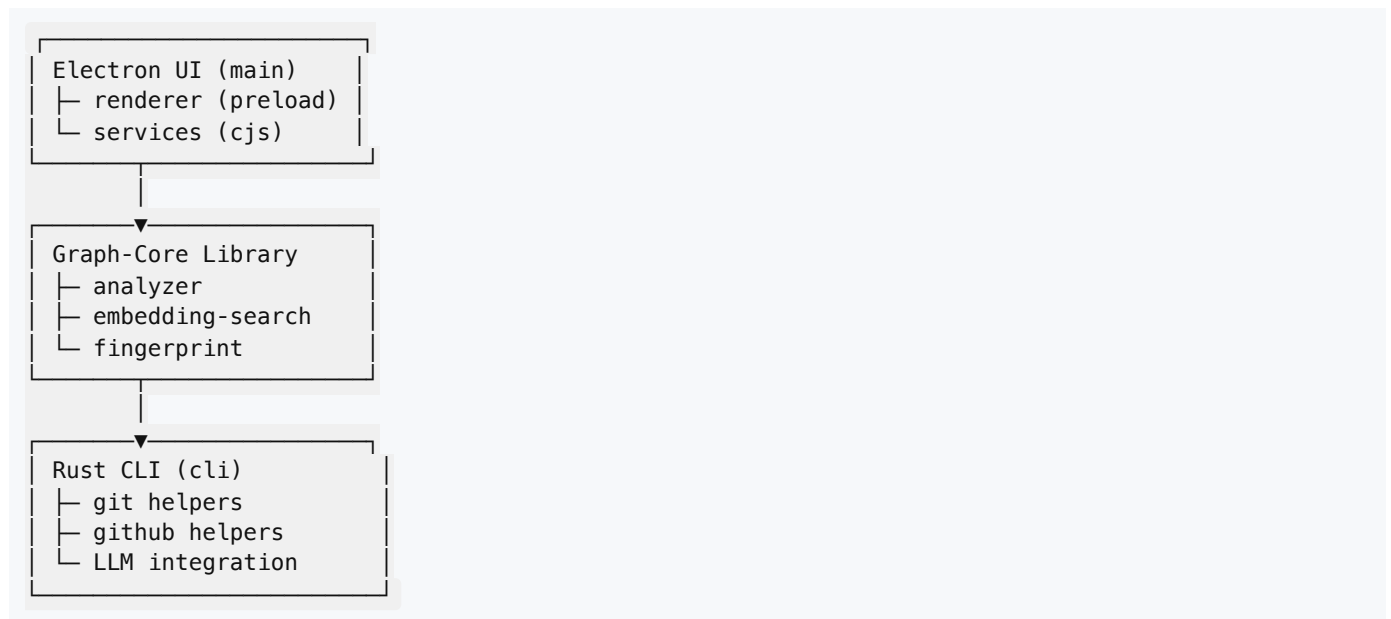
Repository overview

NovaDiff is a cross-platform code-diff analysis tool.

- **Electron UI** – `electron/main.cjs` bootstraps the application and manages windows.
- **Rust CLI** – `cli/src/main.rs` orchestrates Git, GitHub, and LLM calls.
- **Graph-core library** – `packages/graph-core/src` builds and normalises a knowledge graph from repository metadata.

The tool extracts semantic information from a Git repository, builds a graph, and renders visual summaries and documentation.

Architectural layout



Electron services (e.g. `git-service.cjs`, `github-service.cjs`, `llm.cjs`) expose async APIs to the renderer via `preload.cjs`.

The graph-core library is pure TypeScript and is bundled into Electron; the CLI also invokes it through a Node child process.

Key subsystems

Graph-Core (`packages/graph-core`)

- **Analyzer** – builds a node graph from file metadata, detects layers, languages, and semantic concepts.
- **LLM-Analyzer** – crafts prompts, sends them to an LLM backend (Ollama or LM Studio), and parses JSON responses.
- **Fingerprint** – generates content hashes and fingerprints for files, classes, and functions.
- **Embedding-Search** – provides a cosine-similarity engine for semantic search over embeddings.
- **Normalization** – cleans node IDs, infers types, and removes duplicate edges.

Electron UI

- **Main process** (`electron/main.cjs`) – window lifecycle, menu, and IPC handlers.
- **Preload** (`electron/preload.cjs`) – exposes safe APIs to the renderer.
- **Services** (`*.cjs`) – thin wrappers around Git, GitHub, LLM, and graph-core logic.
- **Docs generators** (`novadiff-docs-*.cjs`) – convert graph data into Markdown/HTML/PDF bundles.
- **Prefetch** (`electron/prefetch-summaries.cjs`) – background worker that caches LLM summaries for quick UI display.

CLI (cli/src/main.rs)

- Implements `ChangeKind` enum, path utilities, and change filtering logic.
- Uses `git-service` and `github-service` to gather repository state.
- Calls the graph-core library via a Node child process to build the graph.
- Provides commands for diff comparison, snapshot export, and publishing.

Workspace & Snapshot

- **Workspace-Store** (`electron/workspace-store.cjs`) – persists session state and snapshot metadata.
- **Workspace-History** (`electron/workspace-history.cjs`) – indexes Git commits into snapshot directories.
- **Snapshot-Export** (`electron/export-snapshot.cjs`) – zips a snapshot for sharing or archival.

Dependency signals

- Cross-file calls: ~228 edges, mainly between Electron services and the graph-core library.
- Resolved relative imports: 43 edges, indicating a modular structure.
- External tools: Git, GitHub CLI (`gh`), Node, Rust compiler, LLM backends (Ollama, LM Studio).
- `package.json` lists dependencies such as `electron`, `typescript`, `ts-node`, `@types/node`, `@types/electron`, `rustc`, `cargo`, `gh`, `ollama`, `lm-studio`.

Operational considerations

Area	Notes
Build	<code>npm run build</code> compiles TypeScript, bundles Electron, and builds the Rust CLI (<code>cargo build --release</code>).
Runtime	Requires Node ≥ 18, Git, and <code>gh</code> . An LLM backend must be running locally (Ollama or LM Studio).
Packaging	<code>electron-builder</code> or <code>pkg</code> are used to create platform binaries.
Testing	Jest tests in <code>packages/graph-core/src/__tests__</code> . Rust tests in <code>cli/src</code> .
Performance	Graph building is CPU-bound; LLM calls are async. Prefetch worker mitigates UI lag.
Security	Electron context isolation is enabled; preload exposes only whitelisted APIs.
Extensibility	New language lessons or LLM providers can be added via the analyzer plugin system.

Documentation gaps

- No high-level README for the Electron services; developers must read individual `*.cjs` files.
- CLI documentation is minimal; only the `ChangeKind` enum is described.
- LLM prompt templates are embedded in code; no external docs.
- No diagram of the snapshot directory layout or the workspace history indexing algorithm.
- TypeScript type definitions for the graph-core public API are sparse; consumers rely on inferred types.
- No CI configuration is visible; build scripts are present but not documented.

Suggested onboarding and verification

1. Clone & install

```
git clone https://github.com/yourorg/NovaDiff
cd NovaDiff
npm ci
cargo build --release
```

2. Verify core library

```
npm run test:graph-core
```

3. Run Electron

```
npm start
```

Check that the window opens and the menu appears.

4. Test CLI

```
./target/release/nova-diff --help
```

Run a sample diff command against a local repo.

5. Generate docs

```
npm run docs:generate
```

Inspect the output in `docs/`.

6. Run snapshot export

```
npm run export-snapshot
```

Verify the ZIP contains expected files.

7. Lint & format

```
npm run lint  
npm run format
```

8. **Explore tests** – Graph-core tests in `packages/graph-core/src/__tests__`; Rust tests in `cli/src`. Run them to confirm no regressions.

Target codebase metrics

Target codebase metrics

Repository: NovaDiff

Path: /Users/iamgroot/Documents/GitHub/NovaDiff

Metric	Value
Indexed files	463
Symbol-bearing files	80
Relative import edges	43
Cross-file call edges (heuristic)	228

Top extensions

Extension Count

.ts	210
.tsx	82
.pak	58
.cjs	27
(no ext)	20
.json	14
.plist	9
.css	5
.dylib	4
.mjs	4
.svg	4
.md	3
.yaml	3
.blockmap	2
.html	2
.icns	2
.asar	1
.bin	1
.dat	1
.dmg	1

Top directories

Directory	Files
packages	235
release	103
src	71
electron	27
.	15
.novadiff-graph	4
cli	3
public	2
tests	2
scripts	1

Detected project stacks

- **node:** package.json

Sample files

- .DS_Store
- .gitignore
- .novadiff-graph/config.json
- .novadiff-graph/diff-overlay.json
- .novadiff-graph/knowledge-graph.json
- .novadiff-graph/meta.json
- CONTRIBUTIONS.md
- LICENSE
- README.md
- THIRD_PARTY_NOTICES.md
- cli/Cargo.lock
- cli/Cargo.toml
- cli/src/main.rs
- electron/code-city-model.cjs
- electron/compare-runner.cjs
- electron/diff-excerpt.cjs
- electron/export-snapshot.cjs
- electron/file-summary-export.cjs
- electron/gh-install.cjs
- electron/gh-path.cjs
- electron/git-blame.cjs
- electron/git-commit.cjs
- electron/git-publish.cjs
- electron/git-service.cjs
- electron/github-auth-flow.cjs
- electron/github-commit-context.cjs
- electron/github-service.cjs
- electron/graph/extract-structure.mjs
- electron/knowledge-graph-runner.cjs
- electron/llm.cjs
- electron/main.cjs
- electron/novadiff-docs-html.cjs
- electron/novadiff-docs-pdf.cjs
- electron/novadiff-docs-writer.cjs
- electron/prefetch-summaries.cjs
- electron/preload.cjs
- electron/repo-discovery.cjs
- electron/workspace-files.cjs
- electron/workspace-history.cjs
- electron/workspace-store.cjs
- index.html
- nova-diff-desktop.jpg
- nova-diff-icon.png
- package-lock.json
- package.json
- packages/graph-core/package.json
- packages/graph-core/scripts/generate-alphabetical-catalog.mjs
- packages/graph-core/scripts/language-names.txt
- packages/graph-core/src/__tests__/change-classifier.test.ts
- packages/graph-core/src/__tests__/domain-normalize.test.ts
- packages/graph-core/src/__tests__/domain-persistence.test.ts
- packages/graph-core/src/__tests__/domain-types.test.ts

- packages/graph-core/src/__tests__/embedding-search.test.ts
- packages/graph-core/src/__tests__/fingerprint.test.ts
- packages/graph-core/src/__tests__/framework-registry.test.ts
- packages/graph-core/src/__tests__/ignore-filter.test.ts
- packages/graph-core/src/__tests__/ignore-generator.test.ts
- packages/graph-core/src/__tests__/language-lesson.test.ts
- packages/graph-core/src/__tests__/language-registry.test.ts
- packages/graph-core/src/__tests__/layer-detector.test.ts
- packages/graph-core/src/__tests__/normalize-graph.test.ts
- packages/graph-core/src/__tests__/parsers.test.ts
- packages/graph-core/src/__tests__/plugin-discovery.test.ts
- packages/graph-core/src/__tests__/plugin-registry.test.ts
- packages/graph-core/src/__tests__/schema.test.ts
- packages/graph-core/src/__tests__/search.test.ts
- packages/graph-core/src/__tests__/staleness.test.ts
- packages/graph-core/src/__tests__/tour-generator.test.ts
- packages/graph-core/src/analyzer/graph-builder.test.ts
- packages/graph-core/src/analyzer/graph-builder.ts
- packages/graph-core/src/analyzer/language-lesson.ts
- packages/graph-core/src/analyzer/layer-detector.ts
- packages/graph-core/src/analyzer/llm-analyzer.test.ts
- packages/graph-core/src/analyzer/llm-analyzer.ts
- packages/graph-core/src/analyzer/normalize-graph.ts
- packages/graph-core/src/analyzer/tour-generator.ts
- packages/graph-core/src/change-classifier.ts
- packages/graph-core/src/embedding-search.ts
- packages/graph-core/src/fingerprint.ts
- packages/graph-core/src/ignore-filter.ts

Detected stacks (marker files)

Stack	Markers
node	package.json

Class / symbol index (heuristic)

File	Symbols / defs (regex)
cli/src/main.rs	ChangeKind, path_segments, path_depth, parent_dir, subtree_change_counts, try_build_gitignore_for_root, is_novadiff_docs_reserved_rel, filter_out_novadiff_docs_changes, walk_path_allowed, filter_changes_for_prefetch_gitignore, is_utf8_text_file, is_change_text_eligible, filter_out_non_text_changes, prefetch_summary_queue, empty_summary_evidence, compact_snippet, push_unique_string, line_range_label, detect_file_roles, build_changed_line_ranges, changed_lines_for_side, collect_touched_symbols, is_import_like_line, is_export_like_line, collect_changed_decl_lines, collect_cited_changed_lines, build_verification_hints, build_evidence_badges
electron/code-city-model.cjs	normalizeSide, dominantAuthor, ownershipForRange, selectBlamePaths, buildBlameMap, buildSymbolFileIndex, buildSideNodes, buildCodeCityModelPayload
electron/compare-runner.cjs	cliExecutableName, resolveRustCli, cleanEnv, runCompareEngine, parseEngineStdout, runCompareEngineAsync
electron/diff-excerpt.cjs	buildDiffExcerpt, buildDiffExcerptChunks
electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/file-summary-export.cjs	escapeHtml, escapeMermaidForHtml, injectMermaidFences, markdownToHtml, isReservedNovadiffRel, safeSummarySegments, resolveSummaryDirAbs, safeArtifactSegment, resolveSelectionBaseDirAbs, resolveSelectionDirAbs, buildFileSummaryDocument, buildSelectionSummaryDocument, exportFileSummaryArtifacts, readFileSummaryMarkdowns, exportSelectionSummaryArtifacts, readSelectionSummaryMarkdowns
electron/gh-install.cjs	resolveBrew, getInstallPlan, readGhVersion, getGhToolingStatus, runInstallCommand, installGh
electron/gh-path.cjs	homeDir, extraPathDirs, augmentPathForCli, ghCandidatePaths, resolveViaLoginShell, verifyGh, resolveGhExecutable, clearGhCache, isGhInstalled, ghInstallHint, ghNotFoundError
electron/git-blame.cjs	parseBlamePorcelain, blameFileAtRef, blameFileOwnership
electron/git-commit.cjs	getCommitDetail
electron/git-publish.cjs	buildFullCommitMessage, previewPublish, executePublish
electron/git-service.cjs	gitExecutable, runGit, tryRunGit, isGitRepo, parsePorcelainStatus, getRepoStatus, parseGithubSlugFromUrl, stageAll, stagePaths, unstagePaths, stageDistrict, commit, push, ensureWorktreeBase, removeDirSafe, checkoutRefWorktree, cleanupNovadiffWorktrees, preparePrCompareRoots, detectGitTooling
electron/github-auth-flow.cjs	stripAnsi, cancelGithubDeviceAuth, startGithubDeviceAuth
electron/github-commit-context.cjs	parseJsonSafe, shortGhError, normalizeThread, getGithubCommitContext
electron/github-service.cjs	runGh, tryRunGh, isGhAvailable, getAuthStatus, listRepos, listPullRequests, viewPullRequest, createPullRequest, getUserProfile, listDetectedAccounts, slugFromRepoRoot
electron/graph/extract-structure.mjs	main, buildResult
electron/knowledge-graph-runner.cjs	resolveRepoRoot, resolveNodeExecutable, nodeChildEnv, runCommand, ensureGraphCoreBuilt, loadCore, isTextCandidate, walkProjectFiles, detectLanguage, fileCategory, resolveRelativeImport, buildBatchImportData, runExtractStructureBatch, mapStructuralAnalysis, heuristicComplexity, buildFileMeta, topSegment, buildLayersFromFiles, assignNodeLayers, buildDiffOverlay, getGitHash, buildKnowledgeGraph
electron/llm.cjs	normalizeBase, sendAccumulated, normalizeRepeatUnit, arraysEqual, trimRepeatedParagraphSuffix, trimRepeatedLineSuffix, sanitizeModelResponse, ollamaGenerationOptions, lmStudioGenerationOptions, ollamaNumPredict, streamOllamaChat, lmStudioMaxTokens, streamLmStudioChat, buildWorkspaceDocPrompt, buildCodebaseDocPrompt, buildCommitMessagePrompt, buildSelectionDocPrompt, bulletBlock, renderSummaryEvidenceBlock, buildExplainCodePrompt, buildGroundingContext, verifierWordLimit, buildGroundedRevisionPrompt, shouldRunVerifier, reviseGroundedMarkdown, buildPerFileSummarizePrompt, buildChunkExtractionPrompt, buildChunkSynthesisPrompt

File	Symbols / defs (regex)
electron/main.cjs	sendEngineProgress, getTargetWindow, windowStatePayload, emitWindowState, iconPath, buildMenu, createWindow, sendKnowledgeGraphProgress, sendWorkspaceHistoryProgress, sendGithubAuthProgress
electron/novadiff-docs-html.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, escapeRegex, countMatches, normalizeMalformedCodeFences, looksLikeRawDiffDump, normalizeFenceChunk, normalizeCompactListMarkers, injectMermaidFences, normalizeMarkdownDocument, stripTags, slugify, addHeadingAnchors, markdownToHtml, mermaidSection, navHtml, topDirectory, buildImpactHeatmapHtml, buildSearchIndex, buildEnhancementScript, wrapPage, bundleMetaLine, buildIndexHtml, buildNarrativeHtml, buildMetricsHtml
electron/novadiff-docs-pdf.cjs	bundleKeyOf, bundleBaseDir, bundleModeMeta, escapeHtml, escapeMermaidForHtml, renderSymbolsTable, renderImportEdgesTable, renderDetectedProjects, renderCrossFileCallEdges, mermaidBlock, buildMainPdfHtml, buildDiagramsPdfHtml, waitForPdfReady, htmlToPdfFile, generateNovadiffDocsPdf
electron/novadiff-docs-writer.cjs	bundleKeyOf, bundleBaseDir, bundleHeadline, readme, writeNovadiffDocsBundle
electron/prefetch-summaries.cjs	clearSummaryPrefetchCache, stopSummaryPrefetchWorker, getPrefetchedSummary, defaultPrefetchLimit, runSummaryPrefetchLoop, startSummaryPrefetchWorker
electron/repo-discovery.cjs	homeSearchRoots, readGitRemoteSlug, scanForGitRepos, discoverRepos, matchRepoForGithubRepo
electron/workspace-files.cjs	listSnapshotFiles, readSnapshotTextFile
electron/workspace-history.cjs	removeDirSafe, isUsableSnapshotDir, listCommitsOldestFirst, snapshotCommit, ensureCommitSnapshot, indexWorkspaceHistory, refreshWorkspaceHistory
electron/workspace-store.cjs	sessionPath, workspacesRoot, readJson, withSessionWriteLock, writeJsonAtomic, writeJson, loadSession, setLocalOnlyMode, saveSession, listWorkspaces, upsertWorkspace, setActiveWorkspace, setGitUser, getWorkspace, updateWorkspaceUiState, updateWorkspaceLiveRepo, createWorkspace
packages/graph-core/scripts/generate-alphabetical-catalog.mjs	normalizeKey, slugify, resolveWasmsKey, resolveExtensions, claimExtensions
packages/graph-core/src/__tests__/change-classifier.test.ts	makeAnalysis
packages/graph-core/src/__tests__/plugin-registry.test.ts	createMockPlugin
packages/graph-core/src/__tests__/staleness.test.ts	makeGraph
packages/graph-core/src/analyzer/graph-builder.ts	FileMeta, FileAnalysisMeta, NonCodeFileMeta, NonCodeFileAnalysisMeta, GraphBuilder
packages/graph-core/src/analyzer/language-lesson.ts	LanguageLessonResult, buildConceptPatterns, detectLanguageConcepts, getLanguageDisplayName, buildLanguageLessonPrompt, extractJson, parseLanguageLessonResponse
packages/graph-core/src/analyzer/layer-detector.ts	LLMLayerResponse, toLayerId, matchFileToLayer, detectLayers, buildLayerDetectionPrompt, parseLayerDetectionResponse, applyLLMLayers
packages/graph-core/src/analyzer/llm-analyzer.ts	LLMFileAnalysis, LLMProjectSummary, buildFileAnalysisPrompt, buildProjectSummaryPrompt, extractJson, parseFileAnalysisResponse, parseProjectSummaryResponse
packages/graph-core/src/analyzer/normalize-graph.ts	stripToValidPrefix, normalizeNodeId, normalizeComplexity, DroppedEdge, NormalizationStats, NormalizeBatchResult, inferTypeFromId, normalizeBatchOutput
packages/graph-core/src/analyzer/tour-generator.ts	buildTourGenerationPrompt, parseTourGenerationResponse, generateHeuristicTour
packages/graph-core/src/change-classifier.ts	UpdateDecision, classifyUpdate, detectDirectoryChanges, topDirectory, summarizeChanges
packages/graph-core/src/embedding-search.ts	SemanticSearchOptions, cosineSimilarity, SemanticSearchEngine
packages/graph-core/src/fingerprint.ts	FunctionFingerprint, ClassFingerprint, ImportFingerprint, FileFingerprint, FingerprintStore, FileChangeResult, ChangeAnalysis, contentHash, extractFileFingerprint, compareFingerprints, buildFingerprintStore, analyzeChanges
packages/graph-core/src/ignore-filter.ts	IgnoreFilter, createIgnoreFilter

File	Symbols / defs (regex)
packages/graph-core/src/ignore-generator.ts	parseGitIgnorePatterns, isCoveredByDefaults, generateStarterIgnoreFile
packages/graph-core/src/languages/catalog-builder.ts	catalogEntryToLanguageConfig, buildAlphabeticalLanguageConfigs, listAlphabeticalLanguages
packages/graph-core/src/languages/catalog-types.ts	LanguageCatalogEntry
packages/graph-core/src/languages/framework-registry.ts	FrameworkRegistry
packages/graph-core/src/languages/language-registry.ts	LanguageRegistry
packages/graph-core/src/languages/merge-configs.ts	mergeLanguageConfigs, languageConfigsForRegistry
packages/graph-core/src/persistence/index.ts	ensureDir, sanitiseFilePaths, saveGraph, loadGraph, saveMeta, loadMeta, saveFingerprints, loadFingerprints, saveConfig, loadConfig, saveDomainGraph, loadDomainGraph
packages/graph-core/src/plugins/discovery.ts	PluginEntry, PluginConfig, parsePluginConfig, serializePluginConfig
packages/graph-core/src/plugins/extractors/__tests__/cpp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/csharp-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/go-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/java-extractor.test.ts	parse, Repository
packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	parse, helper, greet, noReturn, noop, multiline, Loggable, Repository, publicFunc, anotherFunc, myFunc, process, main, helper, findUser
packages/graph-core/src/plugins/extractors/__tests__/python-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/ruby-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/__tests__/rust-extractor.test.ts	parse
packages/graph-core/src/plugins/extractors/base-extractor.ts	traverse, getStringValue, findChild, findChildren, hasChildOfType
packages/graph-core/src/plugins/extractors/cpp-extractor.ts	unwrapDeclaratorName, extractFuncDeclName, extractParams, extractReturnType, isStatic, CppExtractor
packages/graph-core/src/plugins/extractors/csharp-extractor.ts	extractParams, extractReturnType, hasModifier, extractUsingSource, lastComponent, extractInvocationName, CSharpExtractor
packages/graph-core/src/plugins/extractors/generic-extractor.ts	nodeName, lineRange, importSource, GenericExtractor
packages/graph-core/src/plugins/extractors/go-extractor.ts	extractParams, extractResultType, extractReceiverType, isExported, GoExtractor
packages/graph-core/src/plugins/extractors/java-extractor.ts	extractParams, extractReturnType, hasModifier, extractScopedIdentifierPath, lastComponent, JavaExtractor
packages/graph-core/src/plugins/extractors/php-extractor.ts	extractParams, extractReturnType, extractUseName, lastSegment, PhpExtractor
packages/graph-core/src/plugins/extractors/python-extractor.ts	extractParams, extractReturnType, unwrapDecorated, PythonExtractor
packages/graph-core/src/plugins/extractors/ruby-extractor.ts	extractParams, extractAttrProperties, getStringContent, RubyExtractor
packages/graph-core/src/plugins/extractors/rust-extractor.ts	extractParams, extractReturnType, isPublic, extractScopedPath, RustExtractor

File	Symbols / defs (regex)
packages/graph-core/src/plugins/extractors/types.ts	LanguageExtractor
packages/graph-core/src/plugins/extractors/typescript-extractor.ts	extractParams, extractReturnType, extractImportSpecifiers, TypeScriptExtractor
packages/graph-core/src/plugins/parsers/dockerfile-parser.ts	DockerfileParser

Module import edges (tabular)

Use together with the flowchart in the Diagrams PDF for a "call graph"-style overview at module boundaries.

From	To
electron/code-city-model.cjs	electron/git-blame.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs
electron/gh-install.cjs	electron/gh-path.cjs
electron/git-commit.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/git-service.cjs
electron/git-publish.cjs	electron/github-service.cjs
electron/github-auth-flow.cjs	electron/gh-path.cjs
electron/github-commit-context.cjs	electron/github-service.cjs
electron/github-service.cjs	electron/git-service.cjs
electron/github-service.cjs	electron/gh-path.cjs
electron/github-service.cjs	electron/github-commit-context.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/main.cjs	electron/llm.cjs
electron/main.cjs	electron/prefetch-summaries.cjs
electron/main.cjs	electron/novadiff-docs-writer.cjs
electron/main.cjs	electron/novadiff-docs-pdf.cjs
electron/main.cjs	electron/novadiff-docs-html.cjs
electron/main.cjs	electron/file-summary-export.cjs
electron/main.cjs	electron/code-city-model.cjs
electron/main.cjs	electron/knowledge-graph-runner.cjs
electron/main.cjs	electron/git-service.cjs
electron/main.cjs	electron/github-service.cjs
electron/main.cjs	electron/git-commit.cjs
electron/main.cjs	electron/repo-discovery.cjs
electron/main.cjs	electron/git-publish.cjs
electron/main.cjs	electron/workspace-store.cjs
electron/main.cjs	electron/git-blame.cjs
electron/main.cjs	electron/workspace-files.cjs
electron/main.cjs	electron/workspace-history.cjs
electron/main.cjs	electron/github-auth-flow.cjs
electron/main.cjs	electron/gh-path.cjs
electron/main.cjs	electron/gh-install.cjs
electron/main.cjs	electron/export-snapshot.cjs
electron/main.cjs	electron/compare-runner.cjs
electron/prefetch-summaries.cjs	electron/llm.cjs
electron/prefetch-summaries.cjs	electron/diff-excerpt.cjs
electron/prefetch-summaries.cjs	electron/file-summary-export.cjs
electron/prefetch-summaries.cjs	electron/git-service.cjs
electron/repo-discovery.cjs	electron/git-service.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
electron/workspace-history.cjs	electron/workspace-store.cjs
packages/graph-core/src/__tests__/ignore-filter.test.ts	packages/graph-core/src/ignore-filter.ts
packages/graph-core/src/__tests__/ignore-generator.test.ts	packages/graph-core/src/ignore-generator.ts

Cross-file call edges (tabular)

Edges appear only when a call site name uniquely resolves to a single definition path (very conservative).

From file	To file	Callee
cli/src/main.rs	electron/git-service.cjs	push
electron/code-city-model.cjs	electron/git-blame.cjs	blameFileOwnership
electron/file-summary-export.cjs	electron/novadiff-docs-html.cjs	normalizeMarkdownDocument
electron/file-summary-export.cjs	electron/novadiff-docs-pdf.cjs	htmlToPdfFile
electron/gh-install.cjs	electron/gh-path.cjs	augmentPathForCli
electron/gh-install.cjs	electron/gh-path.cjs	isGhInstalled
electron/gh-install.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/gh-install.cjs	electron/gh-path.cjs	clearGhCache
electron/git-commit.cjs	electron/git-service.cjs	runGit
electron/git-commit.cjs	electron/git-service.cjs	tryRunGit
electron/git-publish.cjs	electron/git-service.cjs	isGitRepo
electron/git-publish.cjs	electron/git-service.cjs	getRepoStatus
electron/git-publish.cjs	electron/github-service.cjs	getAuthStatus
electron/git-publish.cjs	electron/github-service.cjs	slugFromRepoRoot
electron/git-publish.cjs	electron/git-service.cjs	stagePaths
electron/git-publish.cjs	electron/git-service.cjs	stageAll
electron/git-publish.cjs	electron/git-service.cjs	commit
electron/git-publish.cjs	electron/git-service.cjs	push
electron/git-publish.cjs	electron/github-service.cjs	createPullRequest
electron/github-auth-flow.cjs	electron/gh-path.cjs	isGhInstalled
electron/github-auth-flow.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-auth-flow.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-auth-flow.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-commit-context.cjs	electron/github-service.cjs	getAuthStatus
electron/github-commit-context.cjs	electron/github-service.cjs	tryRunGh
electron/github-service.cjs	electron/gh-path.cjs	resolveGhExecutable
electron/github-service.cjs	electron/gh-path.cjs	augmentPathForCli
electron/github-service.cjs	electron/gh-path.cjs	ghNotFoundError
electron/github-service.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/graph/extract-structure.mjs	packages/graph-core/src/plugins/parsers/index.ts	registerAllParsers
electron/knowledge-graph-runner.cjs	packages/graph-core/src/analyzer/graph-builder.ts	GraphBuilder
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveGraph
electron/knowledge-graph-runner.cjs	packages/graph-core/src/persistence/index.ts	saveMeta
electron/llm.cjs	packages/graph-core/src/plugins/extractors/__tests__/php-extractor.test.ts	process
electron/main.cjs	electron/gh-path.cjs	augmentPathForCli
electron/main.cjs	electron/compare-runner.cjs	runCompareEngineAsync

From file	To file	Callee
electron/main.cjs	electron/compare-runner.cjs	runCompareEngine
electron/main.cjs	electron/llm.cjs	summarizeChange
electron/main.cjs	electron/llm.cjs	summarizeChangeStream
electron/main.cjs	electron/llm.cjs	probeProvider
electron/main.cjs	electron/prefetch-summaries.cjs	stopSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	clearSummaryPrefetchCache
electron/main.cjs	electron/prefetch-summaries.cjs	startSummaryPrefetchWorker
electron/main.cjs	electron/prefetch-summaries.cjs	getPrefetchedSummary
electron/main.cjs	electron/file-summary-export.cjs	exportFileSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readFileSummaryMarkdowns
electron/main.cjs	electron/file-summary-export.cjs	exportSelectionSummaryArtifacts
electron/main.cjs	electron/file-summary-export.cjs	readSelectionSummaryMarkdowns
electron/main.cjs	electron/code-city-model.cjs	buildCodeCityModelPayload
electron/main.cjs	electron/novadiff-docs-writer.cjs	writeNovadiffDocsBundle
electron/main.cjs	electron/novadiff-docs-html.cjs	writeNovadiffDocsHtml
electron/main.cjs	electron/novadiff-docs-pdf.cjs	generateNovadiffDocsPdf
electron/main.cjs	electron/novadiff-docs-html.cjs	refreshNovadiffDocsHtmlFromDisk
electron/main.cjs	electron/knowledge-graph-runner.cjs	buildKnowledgeGraph
electron/main.cjs	electron/git-service.cjs	detectGitTooling
electron/main.cjs	electron/github-service.cjs	getAuthStatus
electron/main.cjs	electron/git-service.cjs	getRepoStatus
electron/main.cjs	electron/repo-discovery.cjs	discoverRepos
electron/main.cjs	electron/repo-discovery.cjs	matchRepoForGithubRepo
electron/main.cjs	electron/github-service.cjs	listRepos
electron/main.cjs	electron/github-service.cjs	listPullRequests
electron/main.cjs	electron/git-service.cjs	isGitRepo
electron/main.cjs	electron/git-service.cjs	preparePrCompareRoots
electron/main.cjs	electron/github-service.cjs	viewPullRequest
electron/main.cjs	electron/git-publish.cjs	previewPublish
electron/main.cjs	electron/git-publish.cjs	executePublish
electron/main.cjs	electron/git-service.cjs	stagePaths
electron/main.cjs	electron/export-snapshot.cjs	exportProjectSnapshotZip
electron/main.cjs	electron/git-service.cjs	stageDistrict
electron/main.cjs	electron/workspace-store.cjs	loadSession
electron/main.cjs	electron/github-service.cjs	listDetectedAccounts
electron/main.cjs	electron/github-service.cjs	getUserProfile
electron/main.cjs	electron/gh-install.cjs	getGhToolingStatus
electron/main.cjs	electron/gh-install.cjs	installGh
electron/main.cjs	electron/workspace-store.cjs	setLocalOnlyMode
electron/main.cjs	electron/github-auth-flow.cjs	startGithubDeviceAuth
electron/main.cjs	electron/github-auth-flow.cjs	cancelGithubDeviceAuth
electron/main.cjs	electron/workspace-store.cjs	setGitUser
electron/main.cjs	electron/workspace-store.cjs	workspacesRoot
electron/main.cjs	electron/git-service.cjs	parseGithubSlugFromUrl
electron/main.cjs	electron/workspace-store.cjs	createWorkspace
electron/main.cjs	electron/workspace-history.cjs	indexWorkspaceHistory
electron/main.cjs	electron/workspace-store.cjs	getWorkspace
electron/main.cjs	electron/workspace-store.cjs	setActiveWorkspace
electron/main.cjs	electron/workspace-store.cjs	listWorkspaces
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceLiveRepo
electron/main.cjs	electron/workspace-store.cjs	updateWorkspaceUiState
electron/main.cjs	electron/git-blame.cjs	blameFileAtRef
electron/main.cjs	electron/git-commit.cjs	getCommitDetail
electron/main.cjs	electron/github-commit-context.cjs	getGithubCommitContext

Codebase outline (JSON excerpt)

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    {
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    },
    {
      "count": 58,
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    {
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    },
    {
      "count": 4,
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    {
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      "ext": ".mjs"
    },
    {
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    {
      "count": 2,
      "ext": ".icns"
    },
    {
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    {
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    },
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```

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  },
  {
    "count": 1,
    "ext": ".zip"
  }
],
"class_diagram_mermaid": "classDiagram\n class CSharpExtractor\n class ChangeAnalysis\n class ChangeKind\n class ClassFingerprint\n class CppExtractor\n class DockerfileParser\n class DroppedEdge\n class EnvParser\n class FileAnalysisMeta\n class FileChangeResult\n class FileFingerprint\n class FileMeta\n class FingerprintStore\n class FrameworkRegistry\n class FunctionFingerprint\n class GenericExtractor\n class GoExtractor\n class GraphBuilder\n class GraphQLParser\n class IgnoreFilter\n class ImportFingerprint\n class JSONConfigParser\n class JavaExtractor\n class LLMFileAnalysis",
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    "via": "push"
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  {
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    "to_file": "electron/git-blame.cjs",
    "via": "blameFileOwnership"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "normalizeMarkdownDocument"
  },
  {
    "from_file": "electron/file-summary-export.cjs",
    "to_file": "electron/novadiff-docs-pdf.cjs",
    "via": "htmlToPdfFile"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "augmentPathForCli"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",

```

```

    "via": "isGhInstalled"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "resolveGhExecutable"
  },
  {
    "from_file": "electron/gh-install.cjs",
    "to_file": "electron/gh-path.cjs",
    "via": "clearGhCache"
  },
  {
    "from_file": "electron/git-commit.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "runGit"
  },
  {
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    "to_file": "electron/git-service.cjs",
    "via": "tryRunGit"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "isGitRepo"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/git-service.cjs",
    "via": "getRepoStatus"
  },
  {
    "from_file": "electron/git-publish.cjs",
    "to_file": "electron/github-service.cjs",
    "via": "getAuthStatus"
  },
  {
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    "to_file": "electron/github-service.cjs",
    "via": "slugFromRepoRoot"
  },
  {
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    "to_file": "electron/git-service.cjs",
    "via": "stagePaths"
  },
  {
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    "to_file": "electron/git-service.cjs",
    "via": "stageAll"
  },
  {
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    "to_file": "electron/git-service.cjs",
    "via": "commit"
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  {
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    "to_file": "electron/git-service.cjs",
    "via": "push"
  },
  {
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    "to_file": "electron/github-service.cjs",
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  },
  {
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    "to_file": "electron/gh-path.cjs",
    "via": "isGhInstalled"
  },
  {
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    "to_file": "electron/gh-path.cjs",
    "via": "ghNotFoundError"
  },
  {
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    "to_file": "electron/gh-path.cjs",
    "via": "resolveGhExecutable"
  },

```

```
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  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-commit-context.cjs",
  "to_file": "electron/github-service.cjs",
  "via": "getAuthStatus"
},
{
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  "to_file": "electron/github-service.cjs",
  "via": "tryRunGh"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "resolveGhExecutable"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "augmentPathForCli"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/gh-path.cjs",
  "via": "ghNotFoundError"
},
{
  "from_file": "electron/github-service.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "parseGithubSlugFromUrl"
},
{
  "from_file": "electron/graph/extract-structure.mjs",
  "to_file": "packages/graph-core/src/plugins/parsers/index.ts",
  "via": "registerAllParsers"
},
{
  "from_file": "electron/knowledge-graph-runner.cjs",
  "to_file": "packages/graph-core/src/analyzer/graph-builder.ts",
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{
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  "to_file": "packages/graph-core/src/persistence/index.ts",
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  "via": "augmentPathForCli"
},
{
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{
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  "via": "runCompareEngine"
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{
  "from_file": "electron/main.cjs",
```

```
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  {
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    "to_file": "electron/llm.cjs",
    "via": "probeProvider"
  },
  {
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    "via": "stopSummaryPrefetchWorker"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "clearSummaryPrefetchCache"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "startSummaryPrefetchWorker"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/prefetch-summaries.cjs",
    "via": "getPrefetchedSummary"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/file-summary-export.cjs",
    "via": "exportFileSummaryArtifacts"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/file-summary-export.cjs",
    "via": "readFileSummaryMarkdowns"
  },
  {
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    "to_file": "electron/file-summary-export.cjs",
    "via": "exportSelectionSummaryArtifacts"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/file-summary-export.cjs",
    "via": "readSelectionSummaryMarkdowns"
  },
  {
    "from_file": "electron/main.cjs",
    "to_file": "electron/code-city-model.cjs",
    "via": "buildCodeCityModelPayload"
  },
  {
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  },
  {
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    "from_file": "electron/main.cjs",
    "to_file": "electron/novadiff-docs-pdf.cjs",
    "via": "generateNovadiffDocsPdf"
  },
  {
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    "to_file": "electron/novadiff-docs-html.cjs",
    "via": "refreshNovadiffDocsHtmlFromDisk"
  },
  {
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    "via": "buildKnowledgeGraph"
  },
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}
```

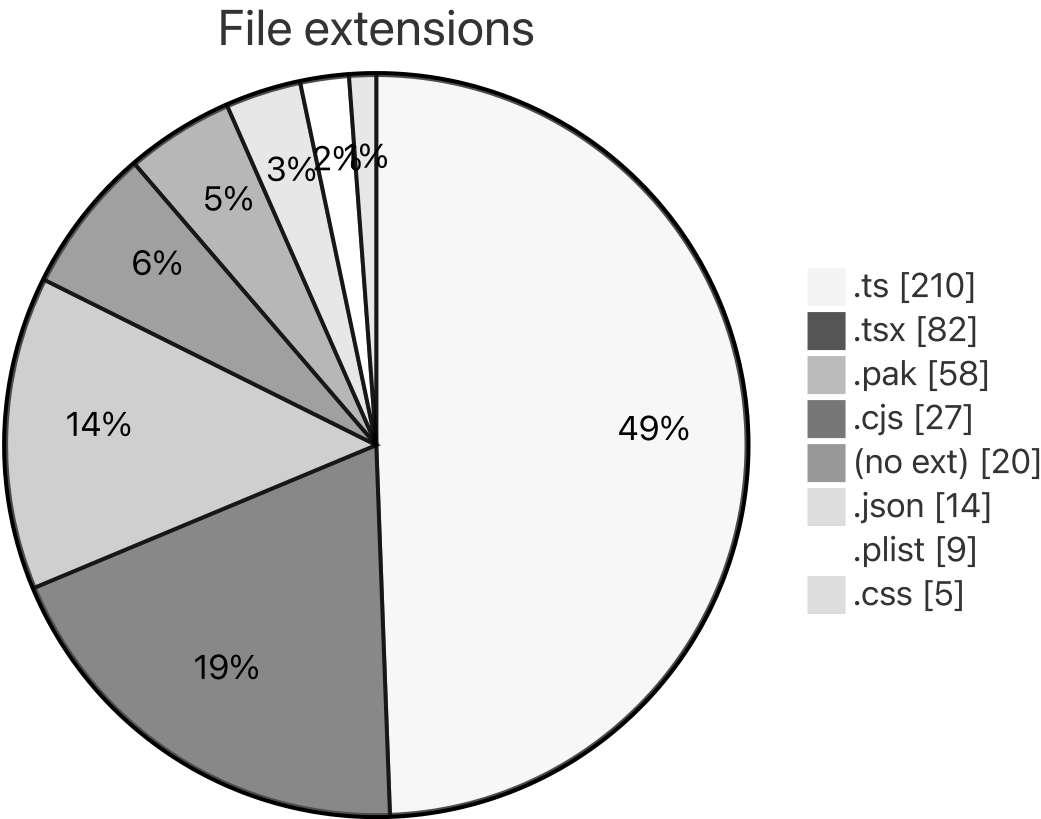
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  "via": "discoverRepos"
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{
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  "to_file": "electron/repo-discovery.cjs",
  "via": "matchRepoForGithubRepo"
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  "to_file": "electron/github-service.cjs",
  "via": "listRepos"
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  "to_file": "electron/github-service.cjs",
  "via": "listPullRequests"
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{
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  "to_file": "electron/git-service.cjs",
  "via": "isGitRepo"
},
{
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  "to_file": "electron/git-service.cjs",
  "via": "preparePrCompareRoots"
},
{
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  "via": "viewPullRequest"
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{
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{
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  "via": "executePublish"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/git-service.cjs",
  "via": "stagePaths"
},
{
  "from_file": "electron/main.cjs",
  "to_file": "electron/export-snapshot.cjs",
  "via": "exportProjectSnapshotZip"
},
{
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  "to_file": "electron/git-service.cjs",
  "via": "stageDistrict"
},
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  "to_file": "electron/workspace-store.cjs",
  "via": "loadSession"
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  "via
... truncated ...

```

Diagrams (Mermaid)

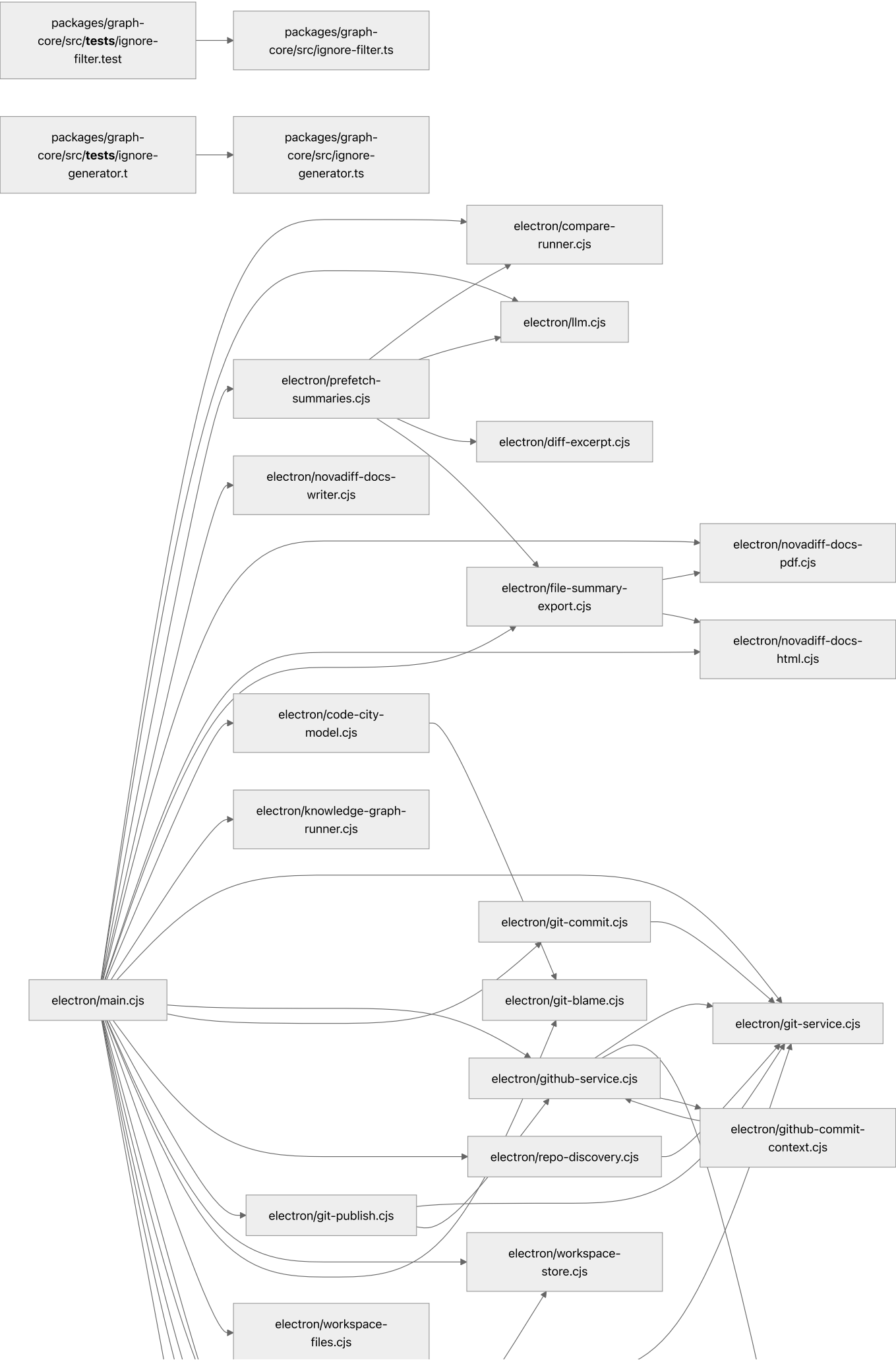
File extension mix

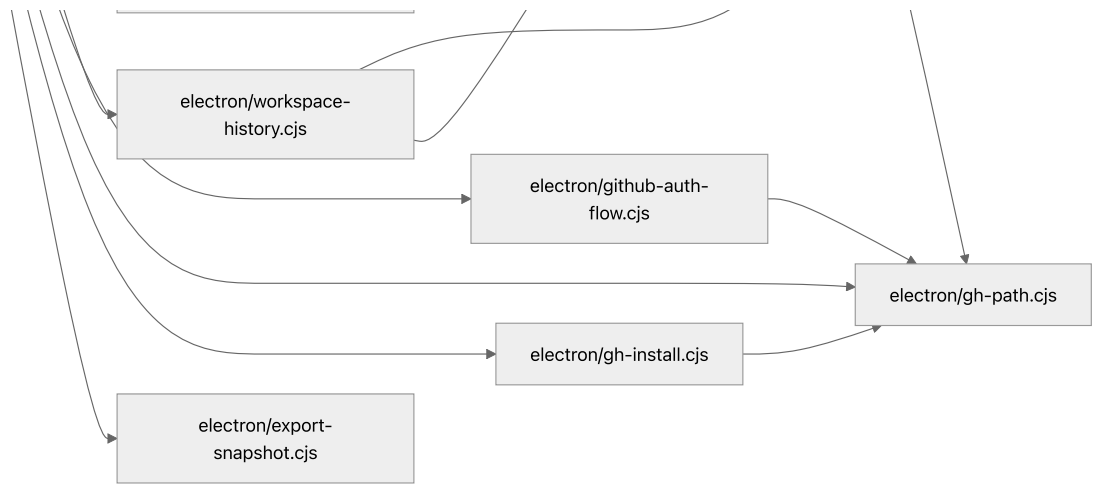


Directory depth

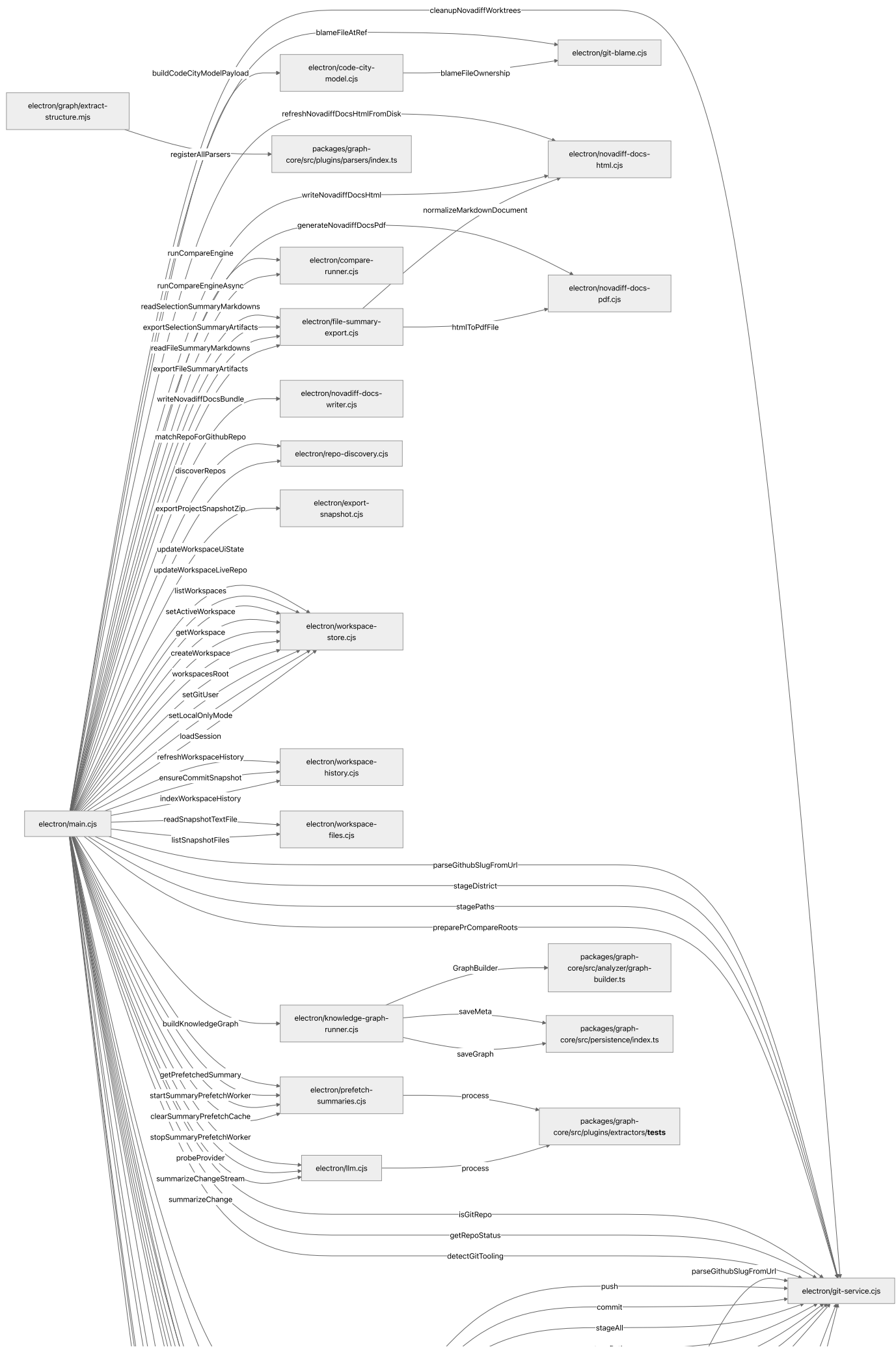


Module import graph (relative imports)





Cross-file calls (heuristic)





CSharpExtractor	ChangeAnalysis	ChangeKind	ClassFingerprint	CopExtractor	DockerFileParser	DroppedEdge	DocParser	FileAnalysisInfo	FileChangeResult	FileFingerprint	FileMeta	FingerprintStore	FrameworkRegistry	FunctionFingerprint	GenericExtractor	GitExtractor	GraphBuilder	GraphQLParser	IgnoreFilter	ImportFingerprint	JSONConfigParser	JavaExtractor	LLVMAnalysis
-----------------	----------------	------------	------------------	--------------	------------------	-------------	-----------	------------------	------------------	-----------------	----------	------------------	-------------------	---------------------	------------------	--------------	--------------	---------------	--------------	-------------------	------------------	---------------	--------------